

TB Vocoder

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Overview

Places the rhythm and articulation of one sound onto the tone of another sound.

What this plug-in solves

Classic vocoders can require a separate synth track and provide little control over intelligibility. TB Vocoder can use its internal carrier for immediate operation or accept a sidechain, while analysis range, envelope timing, imprint, whitening, sibilance, and carrier design remain independently adjustable.

What you can expect

- Classic robotic speech and talk-box tone
- Whisper, Morph, Ring Mod, and Invert transformations
- An internal carrier that moves from a small focused stack to a large cloud
- Detailed control over articulation, band resolution, width, and sibilance

How it works

TB Vocoder takes the changing articulation of one sound and applies it to the sustained tone of another. The modulator supplies rhythm, consonants, and envelope movement; the carrier supplies the pitch and basic color. A bank of frequency areas connects those two roles.

Band and range controls determine how much detail is transferred, envelope controls decide how quickly the carrier follows the modulator, and Imprint or Clarity controls preserve intelligibility. Establish a steady carrier and a clear modulator first, then widen the sound or add character after the words or rhythm are understandable.

Quick start

- 1 Start with Type set to Vocoder, Sidechain off, and a fully wet Mix.
- 2 Begin with a medium Bands setting and choose the useful Range Low and Range High area.
- 3 Tune Attack and Release for intelligibility.
- 4 Adjust Carrier Tune, Count, Color, Resonance, and Width.

- 5** Use Imprint, Detail, Whiten, and Sibilance to clarify speech.
- 6** Try another Type only after the basic carrier works.
- 7** Set Output and final Mix.

Interface and key sections



This is a direct capture of the current plug-in interface. Use the visible labels to locate the areas and controls explained below.

Type

Vocoder applies channel envelopes, Whisper replaces pitch with noise-like energy, Morph blends spectral identities, Ring Mod multiplies structures for metallic sidebands, and Invert creates an inverse spectral relationship.

Bands and analysis range

Bands sets the analysis resolution, while Range Low and Range High restrict the analyzed spectrum. Fewer bands sound classic and coarse; more bands improve intelligibility while using more processing power.

Envelope timing

Attack follows new articulation; Release holds or releases each band. Fast timing is crisp, while slower timing is smoother and more legato.

Internal carrier

Carrier Tune, Resonance, Count, Color, Tone, and Width build the synth source. Count ranges from a small focused stack to a large cloud.

Sidechain

Enable Sidechain to use the external carrier input when the host provides it. Routing must be configured in the DAW.

Imprint and clarity

Imprint sets modulator transfer strength. Detail, Pressure, Whiten, Floor, Sibillance, Warp, and Shift shape clarity, noise floor, formant movement, and articulation.

Mix and Output

Mix blends the transformed result with the original modulator. Output compensates the final level. Programs cover broadcast clarity, robots, choirs, whispers, and broken machines.

Controllable properties

The guide uses the names shown on the plug-in panel. Each explanation describes the audible result, a useful way to approach the control, and an important interaction to listen for.

Control	What it does and when to use it
Attack	Sets how quickly the effect reacts when a sound begins. Faster values control the front; slower values keep more punch. Judge this control while the source repeats in the full arrangement. The right timing should support the groove without making the sound feel detached.
Bands	Sets how many frequency slices carry the speech shape. More bands sound clearer; fewer bands sound more vintage. Move it gradually and listen for the point where the intended result becomes clear without introducing a new problem elsewhere.
Bypass	Turns the complete effect off or on for a direct before-and-after comparison. Compare with bypass at a similar loudness. If the effect creates a tail, let that tail settle before judging the difference.
Carrier Count	Sets how many internal carrier voices are layered together. Move it gradually and listen for the point where the intended result becomes clear without introducing a new problem elsewhere.
Carrier Resonance	Changes the sharpness and tonal focus of the internal carrier voices. Move it gradually and listen for the point where the intended result becomes clear without introducing a new problem elsewhere.
Carrier Tune	Chooses the main note or tonal area used by this control. Choose the musical direction first, then check attacks and sustained notes for unwanted wobble or a change in character.
Color	Changes the overall brightness and tonal character. Sweep broadly to find the useful tonal area, then make the focus narrower and the change smaller while listening in the full mix.
Detail	Sets how much fine speech and transient detail is transferred. Move it until the section reacts too often, then back away until it follows only the events you actually want to control.
Floor	Sets how quiet the input can be before the transferred movement closes down. Move it until the section reacts too often, then back away until it follows only the events you actually want to control.

Control	What it does and when to use it
Imprint	Sets how strongly the articulation and rhythm of the input are printed onto the carrier. Raise it until the contribution is obvious, then reduce it slightly and recheck the sound in the complete mix.
Mix	Blends the original sound with the processed sound. Turn the processed sound up temporarily to learn its character, then return to the mix and keep only the amount that supports the source.
Output	Sets the final listening level after processing. Match the final loudness before deciding whether the processing sounds better. Extra level can make almost any setting seem more impressive.
Pressure	Adds density and firmness to the transferred envelope. Move it until the section reacts too often, then back away until it follows only the events you actually want to control.
Program	Loads a factory starting point. Adjust the controls afterward to fit the current sound. Use a preset as a direction, not a finished answer. Change one section at a time so it is clear which adjustment improves the source.
Range High	Sets the highest frequency used to transfer articulation. Move it gradually and listen for the point where the intended result becomes clear without introducing a new problem elsewhere.
Range Low	Sets the lowest frequency used to transfer articulation. Move it gradually and listen for the point where the intended result becomes clear without introducing a new problem elsewhere.
Release	Sets how quickly the effect relaxes after the sound becomes quieter. Set it in relation to the rhythm and the space between events. Listen for pumping, gaps, or a tail that covers the next sound.
Shift	Moves the selected pitch or frequency structure upward or downward. Choose the musical direction first, then check attacks and sustained notes for unwanted wobble or a change in character.
Sibilance	Restores or emphasizes consonants such as S and T for clearer words. Move it gradually and listen for the point where the intended result becomes clear without introducing a new problem elsewhere.

Control	What it does and when to use it
Sidechain	Uses an external sound as the carrier instead of the internal carrier. Move it gradually and listen for the point where the intended result becomes clear without introducing a new problem elsewhere.
Tone	Makes the vocoder result darker or brighter. Sweep broadly to find the useful tonal area, then make the focus narrower and the change smaller while listening in the full mix.
Type	Chooses the sound character or processing style. Compare the available characters on the same short passage. Choose by the musical result rather than by the name of the mode.
Warp	Stretches or shifts the transferred frequency shape for a different vocal character. Choose the musical direction first, then check attacks and sustained notes for unwanted wobble or a change in character.
Whiten	Evens out the carrier spectrum so speech details transfer more clearly. Move it gradually and listen for the point where the intended result becomes clear without introducing a new problem elsewhere.
Width	Sets the stereo spread of the processed sound. Check the result on speakers, headphones, and in mono when possible. Extreme spatial settings can feel exciting but may not translate everywhere.

Practical uses

Clear robot voice

- Use Vocoder with 32-64 bands.
- Use fast Attack and medium Release.
- Set Imprint high and add Sibilance.
- Use a bright internal carrier with moderate resonance.

Expected result: Speech remains understandable while becoming strongly synthetic.

External musical carrier

- Route a synth to the plug-in sidechain.
- Enable Sidechain.
- Match analysis range to the synth and voice.
- Use Mix at 100 percent on the effect return.

Expected result: The voice articulates the rhythm and harmony of the external carrier.

Common pitfalls

Sidechain availability and bus naming vary by DAW. Return the strongest control toward neutral, compare with bypass at a similar loudness, and add the processing back one section at a time.

More bands are not always more musical. Return the strongest control toward neutral, compare with bypass at a similar loudness, and add the processing back one section at a time.

High carrier counts, width, and resonance can create large peaks. Reduce the main amount, feedback, drive, or input first, then rebuild the sound while watching the output meter and listening after the source stops.